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DEPARTMENT OF PSYCHOLOGY

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ABSTRACTS

1. W. K. Honig & Jeff Willner - Relativistic Determination of the Values of Reinforcement Schedules.

We are teaching pigeons a pair of multiple schedule discriminations in which the more favorable schedule in one discrimination (between VI 90" and VI 270") is the same as the less favorable schedule in the other (between VI 30" and VI 90"). If the value of a schedule, and of the stimulus associated with it, is determined in part by the context of other reinforcement contingencies, then the pigeon ought to prefer the stimulus that signals VI 90" in the former discrimination. We have obtained evidence through several test procedures that supports this expectation.

2. C. A. Barnes - Synaptic Modifiability in Middle Aged and Senescent Rats.

Synapses in middle aged and senescent rats show pronounced differences in their response to repeated presentations of high frequency stimulation. The ability for these synapses to be modified is critically dependent on time of day; however, the circadian cycle of this effect is disrupted in the old animals.

3. S. Nakajima - Endorphinergic neurons and self-stimulation.

Possibility that the endorphinergic neurons may be involved in the generation of rewarding effect was tested using naloxone. Rats with electrodes in the medial frontal cortex, lateral hypothalamus, and ventral tegmentum were trained to press a bar for electrical stimulation. When the response rate became stable, they were intraperitoneally injected with naloxone hydrochloride. Suppression of bar-pressing was observed for some of the electrode placements at relatively high doses of the drug.

4. B. McNaughton - Dissociation of Short and Long Lasting Modification of Synaptic Efficacy at the Terminals of the Perforant Path.

At the perforant path-dentate granule cell synapse, the increased synaptic efficacy following brief episodes of high frequency activation can be resolved into a long lasting component and one or more short lasting components. The long lasting component requires about 50 seconds to develop, following the stimulation, and may last, under some circumstances, for weeks. The short lasting effects are present immediately following activation, and decay completely within minutes. Different rules govern the development of the two effects. They interact in a fashion consistent with the hypothesis that a post-synaptic mechanism underlies the long lasting effect.

5. R. Klein & D. Harding - Hemispheric Processing Efficiency: A 90 Minute Rhythmic Alternation.

Roger Broughton has proposed that the two hemispheres of the brain "take turns" controlling processing. During sleep this alternation is reflected in the cyclic occurrence of rapid eye movement sleep. We are testing performance on two speeded classification tasks designed specifically to tap the processing efficiency of the two hemispheres. Subjects are tested on both tasks every 15 minutes for 8 hours. So far two subjects have been tested in 4 such sessions each. Our results support Broughton's proposal that this rhythm continues during the day-time.

6. J. Stevenson - Differences Between Normal and "Compressed" Tecta Seen by Golgi Methods.

Observations of "compressed" tecta after impregnations by the Golgi-Cox technique suggest an expansion of the dendritic arbor of some tectal elements. Studies in progress using the Rapid Golgi technique to identify the elements involved, and quantitative aspects of the presumed dendritic spread are discussed. Also, the implications of such an expansion are considered.

7. P. Jusczyk & E. Thompson - Perception of a Phonetic Contrast in Multisyllabic Utterances by Two-Month Old Infants.

In the present study, three aspects of the two-month old's perception of multisyllabic utterances were explored. First, do infants perceive phonetic contrasts occurring either in the initial (Bada-Gada) or medial (Daba-Daga) positions of multisyllabic utterances? Second, are infants more likely to perceive these contrasts in stressed as opposed to unstressed syllables? Third, will infants detect a difference between two stress patterns? Our results indicate the following: (1) Two-month olds are sensitive to place-of-articulation differences occurring in either the initial or medial positions of multisyllabic stimuli; (2) Whether the contrast occurred between stressed or unstressed syllables had little or no effect on the infant's ability to detect it; (3) Infants as young as two-months old respond to differences between stress patterns.

8. N. Wintrip - Effects of Monocular Deprivation on Cat Area 17 and 18.

Monocular deprivation has been shown to result in a decrease in cell size of layers of the lateral geniculate nucleus receiving projections from the deprived eye. This effect has been attributed to a differential shrinkage of the Y cell population in this nucleus, the X cells being relatively unaffected. The present study attempts to ascertain whether the Y system is indeed more sensitive to deprivation. X cells project only to area 17 of the cat visual cortex while Y cells project to both area 17 and 18. By comparing the effect of M.D. on areas 17 and 18 this hypothesis may be tested. Preliminary results indicate that monocular deprivation affects areas 17 and 18 equally.

9. B. Timney - Behavioural Evidence for Visual Plasticity in Dark-Reared Cats.

Recovery of visual function was measured behaviourally in 3 cats, 2 raised in darkness until they were 4 months and 1 until it was 6 months old. Initially, the animals appeared to be totally blind, but within about 1 week for the 4-month old cats and about 3 weeks for the 6-month old cat signs of vision were evident. Thereafter, visual acuity, as measured on a jumping stand, progressively improved, approaching normal values within 3 months. These results support recent physiological evidence for recovery of function in the visual cortex following long-term visual deprivation.

10. N. Bennett - Care-Taking and Relatedness in Canids.

Kin selection is a sociobiological model for the evolution of non-parental altruism. Since close relatives share genes, genetic fitness can be increased by promoting the interests of relatives, provided that the cost is sufficiently low. The theory predicts that differential altruism towards relatives will be selected for. In wolves non-reproducing pack members assist in pup care-taking. I will be investigating care-giving in a pack of captive timber wolves and a pack of captive coyote-red wolf hybrids with particular attention to the relatedness of pups and care-givers.

11. K. I. Beverley - Changing Size As a Cue to The Perception of Movement in Depth.

Adapting to a target whose size oscillates about a mean value selectively depresses visual sensitivity to oscillating size. This cannot be attributed to fatigued movement detectors. Furthermore, after adapting to a target whose size increases with a ramp waveform, a subsequently-viewed test target appears to move continuously away in depth. This aftereffect cannot be explained in terms of the classical movement aftereffect. I propose that there are, in the human visual system, detectors selectively sensitive to unidirectionally changing size and which can drive the neural mechanism that underlies the perception of motion in depth.

12. F. H. Harrington - Maintenance of Spatial Organization in Wolf Populations.

Single species often display a remarkable array of social groupings and inter-group relationships as adaptations to variable local ecological situations. In the gray wolf (Canis lupus), some populations are characterized by resident, territorial packs whereas other populations are composed of nomadic packs. Of interest here are the behaviors involved in maintenance of these different pack spatial, land tenure systems. Both scent-marking and howling evidently play important roles in maintenance of territorial spatial systems. An analysis of their respective roles suggests both may also play very similar roles for nomadic land-tenure systems as well.

13. J. Beebe - Serial Processing of Compound Mental Rotations.

Compound mental rotations were made by combining Shepard & Metzler's (1971) two separate rotations of complex abstract figures in picture-plane and depth. Four subjects were each tested on 680 rotations, both single and double, viewed in a tachistoscope. The task was to decide whether a pair of figures were the same or different, and reaction times and errors were recorded. Results showed that the time to perform the double rotations was equal to the sum of the times for the two single rotations. It is concluded that the double operation is done serially rather than in parallel.

14. G. Moran - The Distribution of Social Interactions Among the Member Pairs of a Group of Captive Wolves.

In addition to the analysis of the organisation of movement in the interactions of a group of wolves, some attention has been directed at two higher levels of social behaviour: relationships and social structure. For the past two and one-half years interactions observed during the study have been described and recorded. The patterning of interactions across the member pairs of the group and over time will be described, with particular emphasis placed on "supplanting" interactions. The relevance of this patterning to a meaningful characterisation of social relationships and social structure will be discussed.

15. R. M. Douglas & B. L. McNaughton - Dependence of Enhancement on Stimulus Intensity.

Changes in intensity of electrical pulses applied to the perforant path input to the hippocampus markedly affected the number of presynaptic fibres activated, and the resulting post-synaptic potential. Changes in stimulus strength also affected the amount of enhancement of synaptic efficacy produced by brief high frequency trains. A minimum intensity was needed for enhancement, and higher intensities produced more enhancement. This suggests that enhancement of synapses of one perforant path fibre is conditional on activity in other fibres.

16. B. W. Tansley - Mechanisms of Pattern Vision in Fields of Uniform Luminances.

When an edge is visible in a photic array whose luminance is everywhere the same, there must be a differential stimulation of at least more than one of the three cones in the photopic retina. The question posed in this talk will be: which of the three cone types are involved in the perception of edges (thus forms) when there is equivalent stimulation of their SUM everywhere in the visual display? Two experiments are reported that give strong evidence that only two of the three cone types of the trichromatic eye are necessary for such form perception, and one possible means of carrying out this processing is discussed.

17. M. Cynader - Extension of the Critical Period in Cat Visual Cortex.

It is now generally accepted that response characteristics of single cells in cat visual cortex can be modified by a typical visual exposure only during the first three months of life. Monocular deprivation by eyelid suture results in profound effects on cortical binocular connectivity if performed early in life but has little effect later. We have recorded cortical responses in cats reared from birth to 4-6 months in darkness and subsequently monocularly sutured. The results indicate that the nondeprived eye takes control of the visual cortex in these cats even after the "critical period". Strong effects of monocular deprivation can be obtained in these cats within one week of bringing them into the light.

18. K. Bloom - Factors Which Influence the Demonstration of Infant Vocal Conditioning.

Two groups of infants were identified on the basis of their vocal behavior in the first 2-min baseline period (unresponsive adult) in a social conditioning study. Infants whose vocal rates were less than 2 responses/min were the same infants who looked away from the experimenter's face and who smiled less frequently. Moreover, these low-vocal-rate infants were unaffected by social stimulation. For both groups, baseline vocal rates were indicative of sociability, as measured by smiling, eye contact and vocal responses to social stimulation.

19. M. W. Woolridge - 'Discrete' Behaviour Patterns?

Discrete units of behavioural action have long been the common currency of ethological description and explanation, but there is growing scepticism as to the reality of these units. It has been necessary to attempt to resolve this problem for facial grooming in mice in order to determine the applicability of various behavioural models used to explain the sequential structure of behaviour. The approach has been to quantify the limb and body movements using co-ordinates from high-speed film. The overt behavioural action, in which 'discrete' patterns have been subjectively identified, is then reconstructed from these component actions. When form discontinuities are identified at this level it may then be assessed whether such discontinuities are a function of individual body components, or of their combined action.

20. P. J. Dunham - Changes in Unpunished Behavior During Response Contingent Punishment.

When an operation such as response-contingent punishment suppresses the referent response, how does an organism re-organize the unpunished alternative responses in its repertoire? I will describe some empirically derived rules for predicting the changes in unpunished responding and a recent critical test of these rules.

21. J. Day - Lateral S-R Compatibility in a Simple RT Task.

Three groups of subjects pressed a key with one hand when a stimulus was presented on the right or left of a fixation point. Response hand was varied across blocks of trials, and the response key was positioned on the right, left, or center of the body midline for the three groups. A compatibility effect was found between the position of the stimulus and the ipsilateral hand but not between stimulus position and response key position. Thus the effect depended on the relation between a stimulus and a particular motor output, with facilitation probably depending on intrahemispheric integration of stimulus and response.

22. K. Shapiro - Stimulus Relevance in the Pavlovian Conditioning of Pigeons.

Pigeons were presented with a compound auditory-visual stimulus that terminated with either response-independent electric shock or food. The tone CS was dominant in defensive conditioning, reliably eliciting conditioned prancing. The red light CS was dominant in appetitive conditioning, reliably eliciting pecking at the food hopper. Control groups receiving random presentations of the compound stimulus were run to assess the importance of the forward-pairing procedure in establishing conditioning.

23. M. Connors - Active vs. Passive Exposure: Effects on Kitten Visual Cortex.

Kittens were reared with one eye open when the animal was free to locomote and the other eye open for an equal length of time while the animal was passively restrained in a headholder. Despite equal periods of vision in the two eyes, the "active" eye drove many more cortical neurons than did the "passive" eye. The results indicate that the effects of monocular deprivation by eyelid suture be mimicked simply by preventing visual input to one eye from guiding locomotor behaviour.

24. F. Baker - The Structure of the Normal Goldfish Tectum as Seen by Different Golgi Techniques.

Different aspects of tectum morphology can be illustrated by Golgi-Cox and Rapid Golgi techniques. The Golgi-Cox method shows good overall impregnation of a variety of tectal elements. The various Rapid Golgi techniques, which stain more selectively, show better the detailed structure of individual elements. The results of the two methods combined give a much clearer understanding of the morphology of the tectum.

25. J. Ryon - Den-Digging and Related Behavior in a Captive Timber Wolf Pack.

Timber wolves dig dens annually prior to whelping season in the spring. Den-digging and related behavior of a captive pack of four timber wolves was observed during a single season, encompassing the birth of six pups. Architectural features, method of construction and some aspects of den use not previously reported in the literature are discussed.

26. J. A. McNulty - The Effectiveness of Retrieval cues in the Recall of Large Amounts of Information.

Studies of the effectiveness of retrieval cues upon recall of information have shown that the cues must be present during learning of the information to be effective. Data will be presented to show that large amounts of information can be retrieved using cues not present during learning even when the information to be recalled was presented for only one learning trial.

27. R. Smith - Conditioning Aversions for Prey in Wolves, Coyotes, Hawks.
Produced by: Garcia, Gustavson, Hankins, Kelly, Rusiniak & Brett.

This new film demonstrates use of the taste-aversion paradigm in natural predator control. A cougar refuses to eat venison, a sheep chases hungry wolves, coyotes hide from lambs and rabbits, and hawks decline mice. The film is a must if you attend cocktail parties sponsored by the Wool-growers Association.

28. M. J. Mustari - Effects of Strabismus and Monocular Deprivation on the Developing Kitten Visual Cortex.

Kittens were reared normally until postnatal day 15, when a surgically induced divergent strabismus was induced by section of a medial rectus muscle. These strabismic kittens were allowed normal visual experience until 6 weeks of age when either the strabismic or normal eye was sutured closed for eight days prior to recording from single cells in the striate cortex. It was found that previous strabismus protects neurons in the cortex contralateral (but not ipsilateral) to the eye that had been sutured shut. This seems to hold whether the normal or strabismic eye is the sutured eye. The implications of these results for binocular competition during development and for cortical anatomy will be discussed.

29. J. Barresi - How Fuzzy are the Meanings of Verbal Descriptions?

Several experiments will be reported in which Zadeh's fuzzy set theory is evaluated as a quantitative approach to semantic information processing. Although the results are tentative, they suggest that people may use a combination of logical and fuzzy-logical mental operation in verifying verbal descriptions.

30. J. C. Fentress - Mus Musicus.

As a musical composition integrated patterns of behaviour can be thought of in terms of "notes", "rests", and higher-order "melodies". Among questions of interest is how these levels of order emerge and relate to one another during the lifetimes of individuals. In collaboration with Dr. Ilan Golani the developmental orchestration of some relatively simple motor patterns in mice is used as a model etude. We ask, for example, "do notes" cause "melodies"? Questions about conductors, who translate scores into performance, and of the relations among neuronal instruments, might then be formulated in broader symphonic context.

31. L. Chew - Self-Stimulation in the Neonatal Rat.

Electrical stimulation of the hypothalamus has been found to operantly condition a contact response in rats under 17 days of age. The development and refinement of simple methods of anaesthesia, implantation of electrodes and post-operative care for the neonatal preparation are discussed. The difficulty of selecting an adequate operant response within the neonate's limited repertoire of motor response is considered with regard to the lability, vigour and ease of quantification of such a response. The ontogeny of self-stimulation as measured by this simple technique may clarify the role of the various neural systems which figure in current theories of reinforcement.

32. A. Warner - Adventures in Lower Criminal Courts: Where Might It All Lead?

A discussion of the "whys" behind a naturalistic approach to the study of person-system interactions (the lower criminal courts) relative to past and present laboratory approaches. There will be a brief summary of the methodology and pilot results of court research to date followed by an elaboration of the conceptual framework. Finally, I want to discuss plans for a more comprehensive summer research project specifically focusing on lawyers' influence in sentencing decisions; who has the influence and what is the impact on the defendant?

33. J. Gardner - Space-Time Interactions in Cat Visual Cortex.

It was proposed that binocular cells with interocular latency differences could provide a substrate for a time based stereoscopic depth mechanism. This idea was based on the assumption that simultaneity of binocular input was an important determinant of response strength. Data have now been obtained which justify this assumption and show that single units are highly sensitive to the timing of input from the two eyes. The binocular responses of single units show interactions in both space and time, and suggest that both parameters are used for stereoscopic depth perception.

34. D. M. Regan - New Classes of Neurons in Cat Parastriate Cortex Sensitive to the Direction of Motion in Three-Dimensional Space.

Beverley & Regan suggested on psychophysical grounds that in man there exists a binocular neural mechanism that emphasizes movement in depth, while the well-known stereoscopic mechanism emphasizes position in depth. Cynader & Regan now find in cat visual cortex three types of neuron. Two are tuned to the direction of motion in depth, one for missing, the other for hitting the head: these neurons throw away positional information ("I don't know where it is or what it is, but duck"). The third category may signal positional information in space. The two new cell classes cut across established cell categories (simple, complex, etc.): their behaviour is revealed only by exactly appropriate stimulation. Hitting-the-head neurons occur in clusters-possibly columns.

35. P. J. Urcuioli - Control of Choice Behaviour in Matching-to-Sample by Sample-Specific Behaviours.

Pigeons were trained on matching-to-sample with differential sample-response requirements for producing comparisons. Reversing the sample-response requirements produced i) retarded reacquisition of matching-to-sample but ii) rapid acquisition of oddity-from-sample. Choices between comparisons were apparently controlled by sample-specific behaviours.

36. E. Renner & A. Jusczyk - Some Personal, Social, and Institutional Aspects of Rape.

A woman who has been raped is often doubly victimized: first, by the act; and second, by the legal process, the medical examination, the social embarrassment, and the emotional stress which follows. We will describe an empirical approach for determining the level of social awareness, for understanding the personal stress of the victim, and for identifying the sources of disillusionment with the social and institutional response to rape. The focus of the research was designed to contribute to the public education, the personal counselling, and the social action functions of Halifax Rape Relief.

37. G. V. Goddard - Time and Place: Some Properties of Memory in a Neural Network.

Separate short and long term changes occur at the same synapses in a mammalian control nervous system. They are based on different mechanisms and may have, respectively, pre-synaptic and post-synaptic locations. If all synapses in a theoretical neural net were constructed this way, the error patterns generated during retrieval would differ as a function of time since the learning experience. The differences in error type would be similar in nature to those observed in experiments on human memory. Since correct retrieval (the location of memory) would not differ as a function of time, a single place theory for experience, storage and recall is preserved.

38. I. A. Meinertzhagen - Visual Connections in the Dragonfly.

Dragonflies are active visual predators with remarkably sophisticated vision. Their photoreceptors are arranged in a characteristic tiered fashion within each ommatidium. Each ommatidium projects a set of photoreceptor axons to its own cylindrical cartridge within the first optic neuropile where synaptic associations are established with a small, fixed group of interneurons of several classes. Electronmicroscopic studies reveal the spatial complexity of synaptic interconnexion (dyadic and triadic configurations of postsynaptic elements are found) and the specificity with which particular configurations of elements are associated. These will be illustrated from representative micrographs and data presented to support numerically the range and consistency of synaptic association encountered anatomically.